

⚠ CAUTION

- Do not use a 0.2 mm nozzle. A 0.6 mm nozzle is recommended, followed by a 0.4 mm nozzle.
- It is recommended to use a new nozzle or set the nozzle temperature to 280 °C and clean its interior before printing to ensure it is free of clogs.

Step 1. Dry the filament. For detailed steps, see [Filament Drying](#).

i TIPS

Certain filaments require higher drying temperatures, which the AMS 2 Pro cannot provide. For better drying results, it is recommended to purchase the AMS HT.

Step 2. Apply a layer of solid or liquid glue on the build plate surface.

🔗 NOTE

You can purchase official heatbed solid glue on the Bambu Lab official platform.

Step 3. Preheat the printer.

Before starting the print, set the chamber temperature to the desired value (such as 65 °C) on the printer screen, Bambu Studio, or Bambu Handy. The printer will automatically switch to **heating mode**.

To heat up faster, set the heatbed temperature to 100 °C to assist in raising the chamber temperature and simultaneously preheat the heatbed.

Step 4. Recommended model slicing settings.

- Set the layer height between 0.15 and 0.3 mm, with a maximum print speed of 100 mm/s.
- When printing multiple parts, set the print sequence to By object.
- For large models, choose the Strength preset (6 walls, 25% sparse infill density).

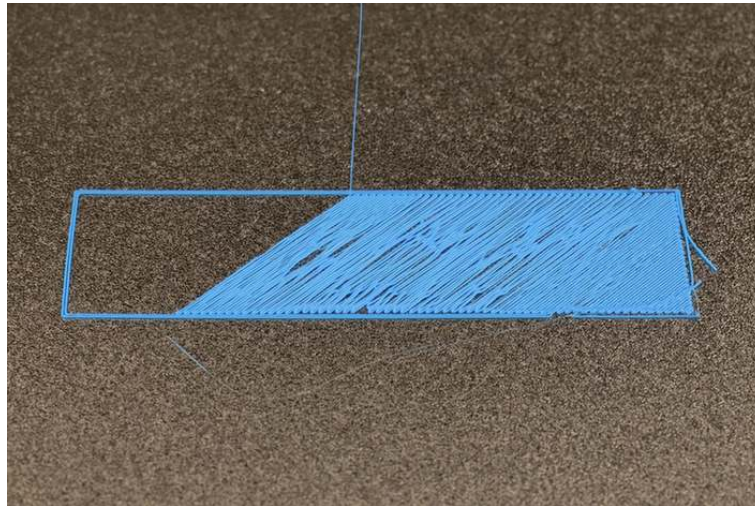
Step 5. After printing, allow the model to cool down to room temperature gradually within the chamber before removal.

i TIPS

If printing carbon-fiber (CF) or glass-fiber (GF) reinforced engineering materials, post-print annealing is recommended. Visit the Bambu Lab Wiki (wiki.bambulab.com/home) and search for "annealing" to find related guidance.

Chapter 9 Print Quality Issues and Solutions

9.1 First Layer Not Sticking



Possible Causes and Solutions

1. Mismatched build plate

Ensure the build plate type matches the filament used, and select the correct build plate in Bambu Studio.

2. Dirty build plate surface

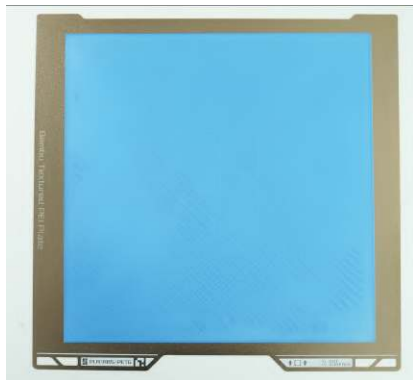
Clean the build plate surface with warm water and a neutral detergent to remove any contaminants without damaging the build plate coat.

3. Bed leveling not performed

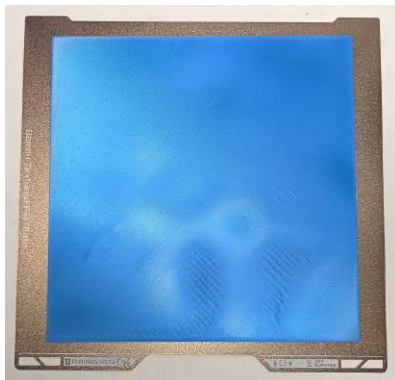
On the printer screen: select  > **Calibration** > **Print Calibration** > **Auto bed Leveling/High-temperature Bed Leveling** to perform bed leveling.

In Bambu Studio, go to the device page, select **Calibration** > **Bed Leveling** to perform bed leveling.

9.2 First Layer Too High/Too Low



Full-plate first layer too low



Partial first layer too low



Full-plate first layer too high

Possible Causes

- The first layer being too low (partial area or full-plate) may cause print failure or damage to the nozzle or build plate.
- The first layer being too high can result in poor adhesion, poor surface quality, or reduced structural strength, and in severe cases, model collapse or shifting, leading to a print failure.

Solutions

- On the printer screen, go to  > **Calibration** > **Print Calibration**, and run Auto Bed Leveling and High-temperature Bed Leveling.
- If the issue persists after bed leveling, visit the Bambu Lab Wiki (wiki.bambulab.com/home) and search for "H2C First Layer" to obtain a detailed tutorial.

9.3 Poor Overhang Quality

When printing overhang structures, if the extruded filament does not cool and solidify quickly, it tends to sag, affecting both appearance and structural integrity.



Possible Causes and Solutions

1. Insufficient support

Check the overhang angle. If it exceeds 45°, support structures should be added.

2. Print speed too high

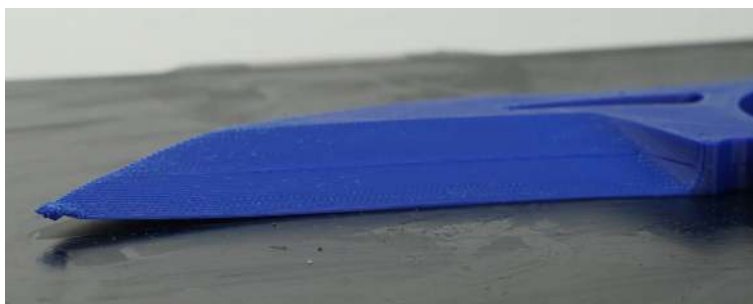
Reduce the printing speed appropriately or enable the Slow down for overhangs function.

3. Insufficient cooling

Lower the nozzle temperature as appropriate, increase the fan speed for the auxiliary part cooling fan and the part cooling fan, and lastly open the front door and top glass cover to help reduce the chamber temperature.

9.4 Model Warping, Falling Off, or Collapse

During printing, localized shrinkage or poor bed adhesion may cause the model to warp, detach, or collapse. Warped areas often exhibit a noticeable horizontal bulge, caused by the nozzle being too close to the warped surface. This results in flattening of the extruded material and subsequent extrusion overflow.



Possible Causes and Solutions

1. Tall model with unstable center of gravity

Add supports, reduce print speed and acceleration. Lay the model flat whenever possible, split the model for printing if necessary.

2. Nozzle collision with the model

Clean the nozzle and slightly increase the nozzle temperature. Reduce print and infill speeds. Choose sparse infill patterns like Gyroid or Concentric to avoid crossing paths.

3. High material shrinkage rate

For large prints, select filaments less prone to warping, such as PLA.

4. Improper printing environment or settings

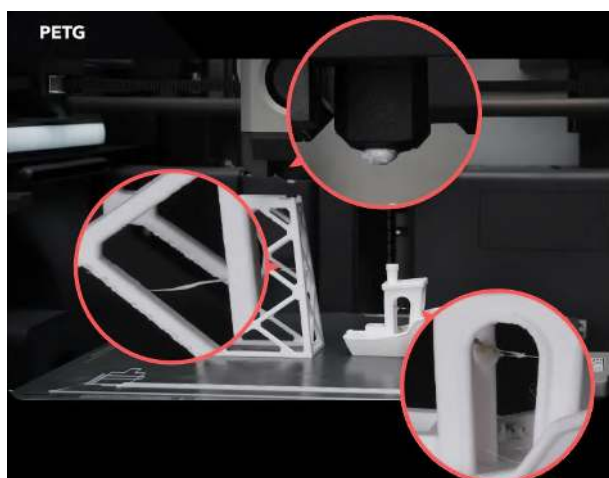
Raise the heatbed temperature appropriately, close the printer's front door and top glass cover, reduce fan speed, lower sparse infill density, and select non-crossing infill patterns like Gyroid.

5. Insufficient bed adhesion

Verify that the build plate information in Bambu Studio matches the actual build plate. Clean the build plate and nozzle. Enable Brim and increase the brim width. Apply glue to the build plate surface if needed to enhance bonding and raise the heatbed temperature slightly.

9.5 Filament Sticking to the Nozzle

During printing, small amounts of molten filament may stick to the nozzle surface. This can cause local under-extrusion, rough surfaces, and—in severe cases—nozzle clog or clumping. This issue is common with high-viscosity filaments such as PETG.



Sticky nozzle



Clean nozzle

Possible Causes and Solutions

1. **Moist filament**

Dry the filament. For detailed steps, see [Filament Drying](#).

2. **Excessive flow rate or worn nozzle**

Adjust the flow rate to reduce over-extrusion, and replace the nozzle if necessary.

3. **Print speed too slow or temperature too high**

Match print speed with nozzle temperature: increase nozzle temperature when printing at higher speed or higher volumetric speed; lower nozzle temperature when printing slowly.

4. **Worn, loose, misaligned, or damaged nozzle wiping part**

Identify issues by visually inspecting, manually moving the wiping part, or pushing the tool-head against it after powering off. Repair or replace it if abnormalities are found.

5. **Too many travel paths in the model**

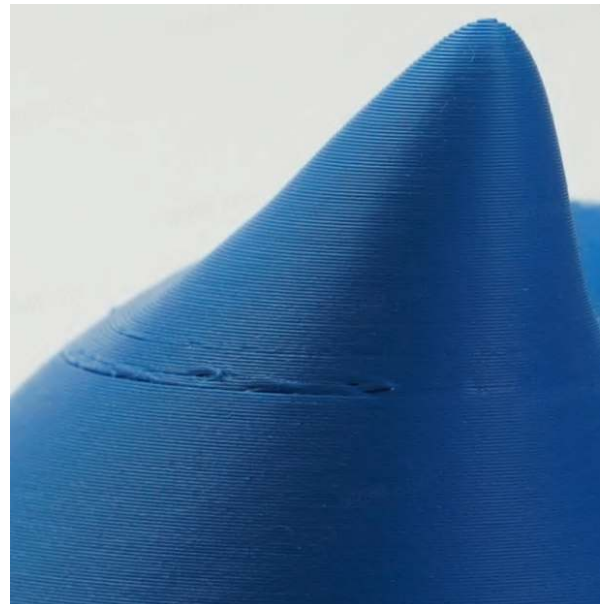
Optimize the print layout to avoid excessive travel paths between model sections.

9.6 Under-Extrusion

Under-extrusion occurs when the printer does not extrude enough filament, causing incomplete infill or surface discontinuities. It can present as **overall under-extrusion** or **partial under-extrusion**.



Full-plate under-extrusion



Partial under-extrusion

Possible Causes and Solutions

1. **Excessive extrusion resistance**

Check whether the spool rotates freely or if the filament is tangled. Clean or replace the PTFE tube, clean or replace the extruder gears, and unclog or replace the nozzle.

2. **Insufficient extrusion**

Slightly increase the nozzle temperature or reduce the print speed. Unclog or replace the nozzle, and adjust the flow ratio in Bambu Studio accordingly.

3. **Under-extrusion at corners**

This may be due to an incorrect pressure advance value, requiring a recalibration of the flow rate.

9.7 Stringing and Oozing

Stringing and oozing typically appear as thin filament strands or small blobs on the surface. It is commonly caused by excessive extrusion, moist filament, or abnormal melting and flow behavior of the filament.



Stringing model



Normal model

Possible Causes and Solutions

1. **Moist filament**

Dry the filament. For detailed steps, see [Filament Drying](#).

2. **Long travel distance with small retraction length**

Increase the retraction length or speed appropriately to prevent molten filament from leaking during travel moves.

3. **Special model structure or improper placement**

Reduce the spacing between models and enable Avoid crossing walls feature in Bambu Studio.

4. **Nozzle temperature too high**

Lower the nozzle temperature moderately to increase filament melt viscosity, making it less "liquid" and reducing oozing.

5. **Using low-density filaments (such as PLA Aero) without reducing temperature or flow rate**

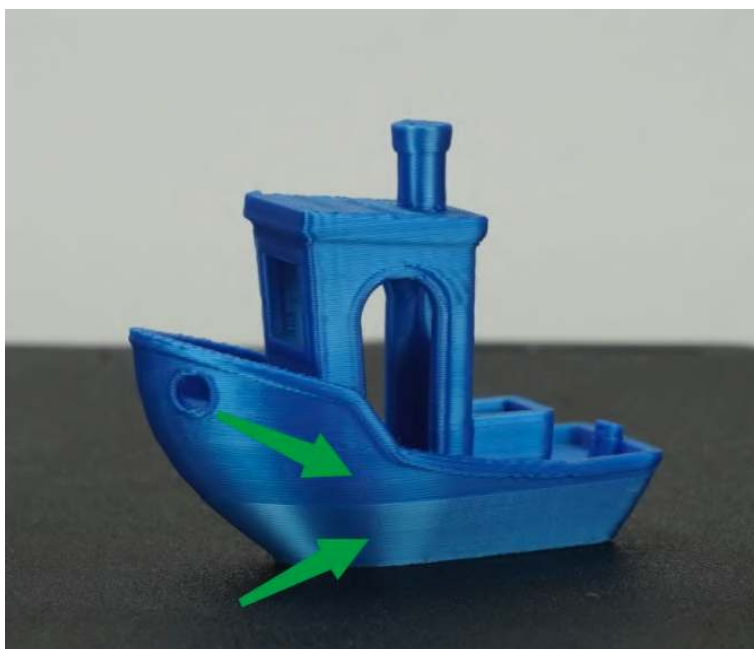
Lower the print temperature appropriately and adjust the flow rate to the range of 0.5 – 0.7.

6. **Worn or oversized nozzle**

Ensure the actual nozzle matches the slicing settings. Check the nozzle condition and replace it if worn.

9.8 Gloss Difference

At the same temperature, a slower print speed produces a smoother surface. At the same print speed, a higher temperature results in a glossier surface finish. If uneven surface gloss appears during printing, it is usually caused by differences in material melting behavior or cooling rates, which lead to variations in flow leveling and surface roughness. This phenomenon is more noticeable with highly reflective materials.



Possible Causes and Solutions

1. **Significant differences in print speed across different areas of the model (such as overhang areas)**

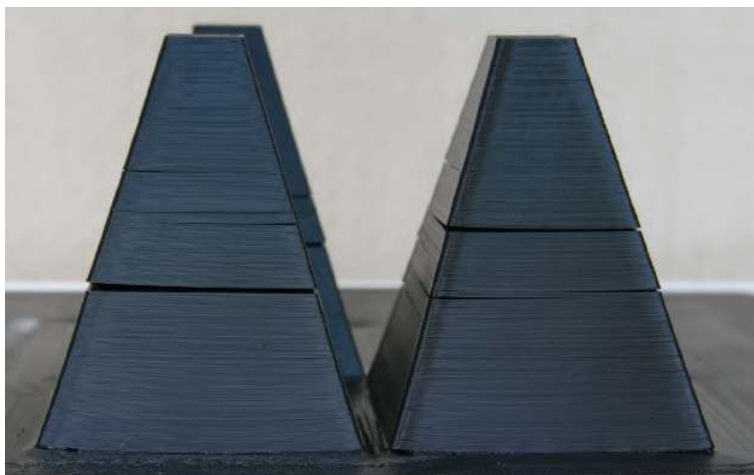
For large models, slightly reduce outer wall speed; for small models, slightly reduce overall print speed. If necessary, disable Slow down for overhangs.

2. **High print speed when using a low layer height (such as 0.08 mm)**

This may cause a fish-scale effect on the model surface. Consider increasing the layer height, lowering print speed, or raising print temperature accordingly.

9.9 Interlayer Cracking

Interlayer cracking occurs due to insufficient bonding between layers, resulting in visible cracks on the model surface. This issue is commonly seen when printing ABS, ASA, PC, PET-CF, PA-CF, and high-shrinkage filaments.



Possible Causes and Solutions

1. **Under-extrusion leading to gaps between lines**

Clean and unclog the nozzle, appropriately increase the nozzle temperature, or reduce the print speed.

2. **Weak interlayer adhesion or thin structural sections**

Increase the number of model wall loops or raise the sparse infill density to enhance strength.

3. **Excessive cooling**

Reduce fan speed appropriately, increase the bed temperature, and close the printer's front door and top glass cover.

9.10 Seam

In FDM 3D printing, the transition point where each layer starts and ends naturally forms a seam. The following methods can be used to optimize the seam appearance and improve surface quality.



Optimization Methods

1. Set the number of wall loops to 3.
2. Print the model individually to avoid seam accumulation from multi-model prints.
3. Appropriately increase the nozzle temperature and reduce the outer wall speed.
4. For ring-shaped or rotationally symmetric models, try enabling **Spiral vase** mode.

9.11 Belt Pattern

During printing, contact between the belt and idler pulley may leave stripes on the model surface, matching the belt tooth pitch (2 mm). This is a common occurrence when using CoreXY 3D printers.



Solutions

1. Increase the outer wall print speed to 200 mm/s.
2. If the filament's default speed is low, try reducing the layer height or slightly increasing the temperature and the maximum volumetric speed.

9.12 Top Layer Gaps

Due to incorrect flow ratio settings, nozzle issues, or unstable extrusion, noticeable gaps may appear on the model's top layer.



Possible Causes and Solutions

1. Incorrect nozzle size

Verify that the nozzle diameter set in the slicing software matches the actual nozzle.

2. Extrusion blockage

Clean and maintain the extruder, PTFE tube, and nozzle.

3. Improper flow rate (too high or too low)

Recalibrate the flow rate on the Calibration page of Bambu Studio.

Chapter 10 Other Common Issues and Solutions

10.1 Printable Area Error

When Bambu Studio detects that the placement of a model conflicts with the printable area, a warning or error message will be triggered and displayed. For more information about the print area, see [Check the Model Placement Area](#).

Possible Causes and Solutions

1. **The model's default filament is in the non-printable area of the corresponding nozzle.**

Adjust the model position to ensure it is within the printable area of the assigned nozzle.

2. **The embedded part of a model's printed area is in the non-printable area of a certain nozzle.**

After slicing, check the internal layer position and adjust the model so that its internal layers are within the corresponding printable area of the corresponding nozzle.

3. **Flush to object's support/infill is enabled, causing the G-code path to exceed the printable area of a certain nozzle.**

Adjust the position of the support or infill so that it is fully within the printable area of the assigned nozzle.

4. **The support filament is in the non-printable area of the corresponding nozzle.**

Adjust the model position to ensure all supports are within the printable area of the assigned nozzle.

10.2 Nozzle Offset Calibration Failure

The causes of nozzle offset calibration failure and high-precision nozzle offset calibration failure are different. Please troubleshoot according to the following common causes and corresponding solutions.

Possible Causes and Solutions for Nozzle Offset Calibration Failure

1. **Calibration sensor is not detected or large positioning deviation**

Check for related HMS error messages displayed by the printer.

2. **Non-official hotend installed**

Replace with an official hotend.

3. **Hotend not properly installed**

Check if the hotend clip is securely fastened.

Possible Causes and Solutions for High-Precision Nozzle Offset Calibration Failure

1. **Poor print quality of the calibration model**

Remove existing calibration prints, clean the build plate, and rerun bed leveling. Check the nozzle for clogging or oozing, and clean or replace it if necessary.

2. **Low color contrast between the two calibration filaments**

Use two filaments with a higher color contrast for calibration.

3. **LED lights were not enabled during calibration**

Turn on both LED lights and keep them on throughout the calibration process.

4. **Toolhead camera lens is dirty**

Clean the lens with alcohol wipes.

5. **Abnormal toolhead camera installation (tilted or damaged)**

Reinstall or replace the toolhead camera.

10.3 Motion Accuracy Calibration Failure

If the motion accuracy calibration fails, the printer will display an error message and disable the motion accuracy enhancement feature.

Possible Causes and Solutions

1. **Toolhead camera is dirty or not functioning**

Check if the toolhead camera lens is clean or has any dirt or smudges. If so, gently wipe with lens tissue and also check for related HMS errors.

2. **Incorrect build plate type or vision encoder not placed correctly**

Use the official vision encoder and ensure it is correctly placed on the heatbed.

3. **Vision encoder pattern is dirty or damaged**

Wipe the top layer of the vision encoder with alcohol and inspect for obvious damage or deformation. If any is found, replace the vision encoder.

4. **"Calibration Failed" shown after calibration completes**

This indicates the post-calibration motion accuracy does not meet expectations and requires recalibration.

5. Chamber LEDs or external lights enabled during calibration, causing camera overexposure

Turn off the chamber LEDs and any external light sources, then retry the calibration.

10.4 Clogging Troubleshooting

If you experience the following issues during printing, please follow the troubleshooting steps to diagnose the problem.

- **No filament or very little extrusion:** The extruder gear rotates normally, but no filament comes out of the nozzle, or the extruded filament is extremely thin.
- **Clicking noises from the extruder:** This sound occurs when the extruder servo motor slips because it cannot push filament into the hotend.
- **Under-extrusion or layer gaps in the print:** Layers do not bond properly, leaving visible gaps or holes.
- **Print fails immediately:** The toolhead moves, but no filament is extruded from the nozzle.

Possible Causes

1. Excessive resistance inside the PTFE Tube, preventing smooth filament loading.
2. Clogging inside the extruder, including the dual extruder idlers, extruder gear, the gap between them, and the extruder filament guide.
3. Clogged nozzle or hotend throat, preventing proper melting or extrusion.

Solutions

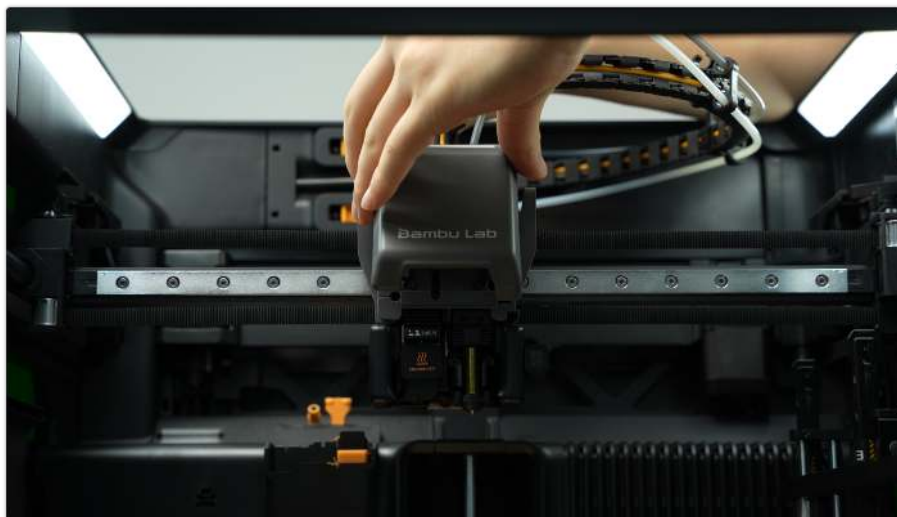
Please see H2C Clog Inspection (wiki.bambulab.com/h2c/troubleshooting/clogging) for detailed troubleshooting steps and solutions. Below is an overview of the procedures.

NOTE

The following instructions use the right hotend as an example. The left hotend uses the same procedure; simply switch to the corresponding left-side components.

Step 1. Remove the toolhead enhanced cooling fan

Open the printer's front door and remove the top glass cover. Unplug the toolhead enhanced cooling fan connector, pinch the top of the fan, and lift it up to remove it.



Step 2. Remove the induction hotend

On the printer touchscreen, go to **⚙️ > Motion: XYZ**, tap **▼** to lower the heatbed, then tap **-Y** to move the toolhead forward for easy access.

On the printer touchscreen, tap **⚙️ > Nozzle & Extruder**, confirm the nozzle temperature has cooled to room temperature, then press the Filament cutter lever to cut the filament. Tap **Right** to move the flow blocker and expose the right hotend.

Pull the induction hotend latch to the right to unlock it, and gently remove the Vortek induction hotend by holding the nozzle tip and pulling diagonally.



NOTE

To remove the left hotend, first cut the left-side filament, remove the silicone sock, unlock the one-clip mechanism, then remove the left hotend.

⚠ CAUTION

Ensure the nozzle is at room temperature before cutting filament to avoid secondary clogs.

Step 3. Manually unload the filament

Press the black outer ring and remove the PTFE Tube from the toolhead.

Push the right idler arm forcefully to the right and hold it in place, then gently pull the filament up with the other hand.

- If filament can be smoothly removed, proceed to Step 4.
- If you feel strong resistance, do not force it. Skip to Step 6 for extruder clog troubleshooting and cleaning.

**⚠ CAUTION**

Forcing filament out when there is resistance may jam it in the dual extruder idlers, making later cleaning more difficult.

Step 4. Manual extrusion

Insert a new filament into the right toolhead filament inlet and try manual extrusion.

On the printer touchscreen, tap **⚙ > Nozzle & Extruder > Extruder**, select **Right**, then tap **▶** to extrude filament. If a prompt appears stating “Please heat the nozzle to above 170 °C”, you may select “**Don't remind me**”.

- If filament extrudes smoothly, clogging may be in the nozzle. Proceed to Step 5 for clog cleaning.

- If filament cannot be extruded or retracted, the extruder may be jammed. Skip to Step 6 for extruder clog troubleshooting and cleaning.

Step 5. Clear hotend/nozzle clogs

Pinch the induction nozzle end and place it onto the right hotend. Push the induction hotend latch to the left until it cannot move further, and gently shake to confirm secure installation.

On the touchscreen home interface, go to  > **Nozzle & Extruder**.

Select **Right** (for right hotend), then set the hotend temperature slightly above the filament's normal print temperature (such as PLA: 220 °C).

Once heated, slowly insert the unclogging pin into the nozzle and move it up and down several times to clear the clog.



DANGER

Filament may unexpectedly eject due to internal pressure release. Always wear heat-resistant gloves during cleaning, and keep your head and body away from the printer chamber to avoid burns.

TIPS

For more clearing methods, refer to Nozzle/Hotend Unclogging Guide.

Step 6. Clear extruder clogs

If manual filament unloading or extrusion fails, clogs may be:

- Filament stuck in the gap between the extruder gear and the dual extruder idlers.

- Filament stuck at the extruder filament guide entrance or inside the guide.
- For the right hotend, filament stuck at or inside the idler arm exit.

For the above cases, disassemble the filament cutter, extruder filament guide, and extruder dual idlers as needed, then remove any stuck filament.

 NOTE

Extruder cleaning is complicated. Please see H2C Clog Inspection (wiki.bambulab.com/h2c/troubleshooting/clogging) for detailed instructions.

10.5 Nozzle/Hotend Unclogging Guide

Hotend clogging is relatively common in FDM 3D printing and can present in the following ways.

- **Under-extrusion:** Insufficient filament extrusion, leading to gaps or layer breaks on the model surface.
- **No extrusion:** The printer operates normally, but no filament is extruded from the nozzle.

Possible Causes

1. **Heat creep softening:** When printing low-temperature materials (such as PLA) in an enclosed environment, the filament may soften before entering the hotend.
2. **Abnormal filament diameter:** The filament diameter is uneven or exceeds 1.75 mm, causing jams at the hotend entrance.
3. **Foreign object clogging:** Extruder debris or filament fragments can enter the hotend, causing partial or complete nozzle blockage.
4. **Particle-filled filaments:** Filaments containing particles (such as carbon fiber-reinforced, glow-in-the-dark, or glitter filaments) may accumulate at the nozzle tip and cause clogs.
5. **Material residue from mixed filaments:** When switching between filament types (such as PLA to PC, ASA to TPU), leftover filament may not be fully purged, leading to mixing and clogging.

Solutions

There are 4 common methods for unclogging a nozzle. Choose the one that best suits your situation.

1. **Manual extrusion:** Heat the nozzle and manually push the filament to extrude the clogged portion.
2. **Unclogging pin cleaning:** Use the unclogging pin to clear the nozzle tip.

3. **Cold pull:** Heat the nozzle, then cool it to room temperature and quickly pull out the filament to remove the blockage.
4. **Hot Allen key cleaning:** Heat the nozzle and insert an Allen key into it to clear residual filament. This is suitable for cold-end clogs.

10.5.1 Manual Extrusion

Step 1. Heat the hotend. On the printer screen, tap  > **Nozzle & Extruder**, then select the left or right hotend that needs cleaning. Set the hotend temperature slightly above the recommended printing temperature for your filament (such as set PLA to 220 °C), then tap Confirm to start heating.

Step 2. Manually extrude. When the temperature reaches the set value, on the printer touch-screen navigate to  > **Nozzle & Extruder**, then tap  at the corresponding nozzle to extrude filament and check if the nozzle extrudes properly.

If the filament cannot be extruded or does not form a continuous strand, proceed to Unclogging Pin Cleaning (see step 2).

TIPS

When using TPU, tap the load button no more than three times. Do not rapidly or repeatedly tap the load/unload button, as this may cause clogging or filament entanglement.



10.5.2 Unclogging Pin Cleaning

CAUTION

This method does not apply to nozzles smaller than 0.4 mm.

- Step 1. Heat the hotend. On the printer screen, select  > **Nozzle & Extruder**, then choose the left or right hotend that needs cleaning. Set the hotend temperature slightly above the recommended printing temperature for your filament (such as set PLA to 220 °C), then tap Confirm to start heating.
- Step 2. Once the hotend reaches the set temperature, insert the unclogging pin into the nozzle and move it up and down several times.

 DANGER

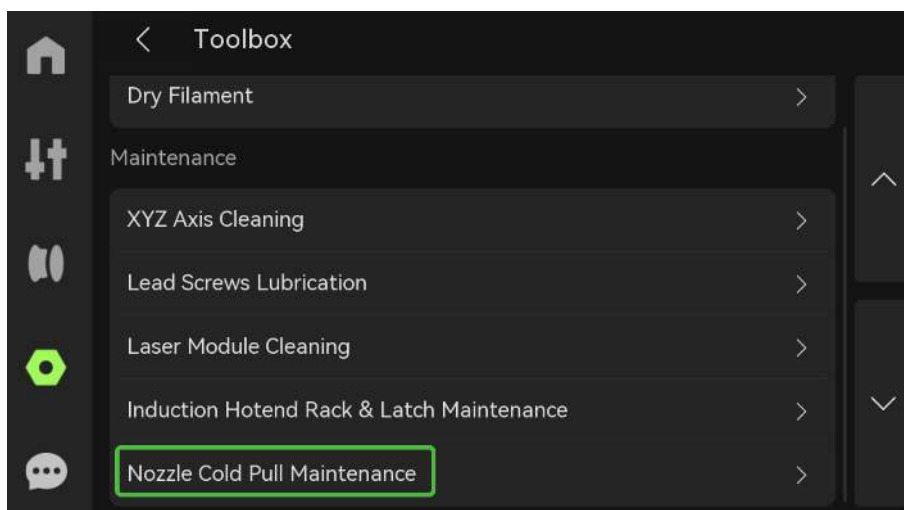
Sudden pressure release inside the nozzle may cause the hot filament to eject unexpectedly. Always wear protective gloves during this process, and keep your hands and body away from the hotend assembly.



10.5.3 Cold Pull

"**Cold pull**" is a 3D printer maintenance technique used to remove debris, clogs, or residual filament from inside the nozzle.

On the printer screen, go to  > **Toolbox** > **Nozzle Cold Pull Maintenance**, then follow the on-screen instructions to complete the process.



You will be guided through the following steps:

- Step 1. Select the hotend and filament that need cleaning.
- Step 2. Remove the PTFE tube from the toolhead.
- Step 3. Insert the previously used filament. The printer will automatically heat the nozzle, purge residual filament, and cool it to the proper temperature.
- Step 4. Remove the filament and reinstall the PTFE tube.

10.5.4 Hot Allen Key Cleaning (Left Hotend)

CAUTION

This method only applies to Allen key H1.5.

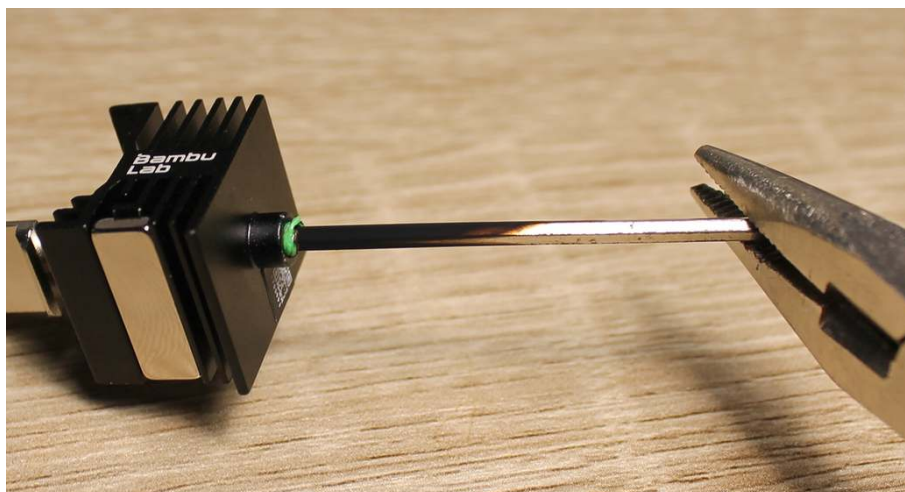
DANGER

Wear protective gloves while operating to prevent burns from dripping or molten filament.

- Step 1. On the printer touchscreen, tap  > **Nozzle & Extruder** > **Extruder**, select **Left**, then remove the silicone sock, unlock the nozzle clip, and remove the left hotend.
- Step 2. Use pliers to hold the Allen key securely and heat it for approximately 10 seconds.



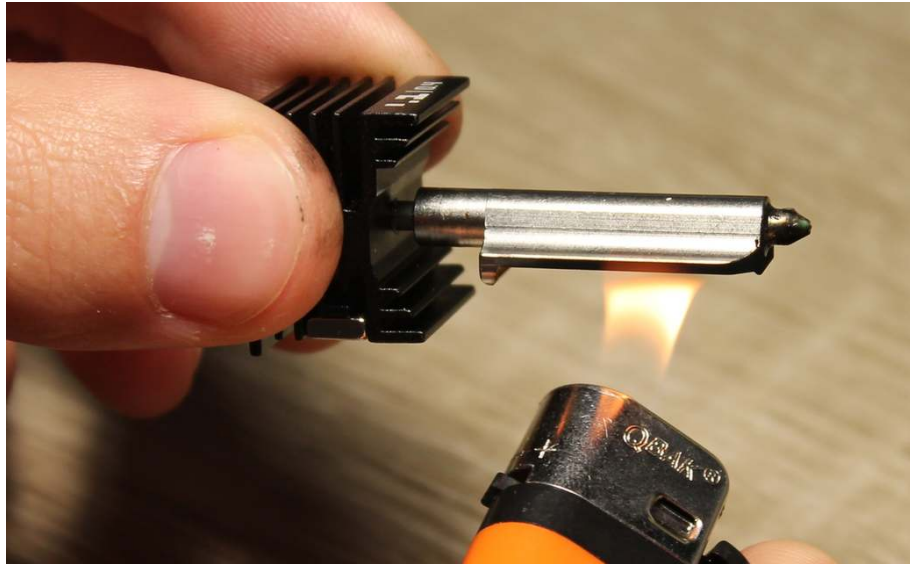
Step 3. While the tip of the Allen key is still hot, quickly insert it into the hotend through the heatsink, pushing until it passes through the filament inside. Then wait about 30 seconds for the Allen key to cool slightly.



Step 4. Now the Allen key is in the hotend, surrounded by filament. Use a standard gas lighter to heat the nozzle tip for about 20 seconds.

⚠ DANGER

- Do not overheat the nozzle or use high-power butane torches. Only use a standard gas lighter.
- The nozzle tip just needs to be hot enough that you can remove the Allen key along with the filament inside.
- Overheating the nozzle may cause filament to drip or splatter. Always keep the nozzle tip pointed away from yourself.



Step 5. After heating, gently pull out the Allen key. The filament and clog material should come out in one piece with the key.



Step 6. Once finished, reinstall the hotend.

10.5.5 Hot Allen Key Cleaning (Right Hotend/Induction Hotend)

DANGER

Always wear protective gloves while operating to prevent burns from dripping or molten filament.

CAUTION

This method only applies to Allen key H1.5.

⚠ CAUTION

Avoid exposing the induction hotend magnet to high temperatures, strong magnetic fields, or severe impacts, as this may cause demagnetization and affect proper function.

- Step 1. On the printer touchscreen, select **⚙ > Nozzle & Extruder > Extruder**. Choose **Right** and set the right hotend temperature slightly above the filament's normal print temperature (such as 220 °C for PLA).
- Step 2. Then, on the printer touchscreen, select **Hotends & Rack**. Tap the induction hotend on the toolhead, then choose **Nest**. The printer will automatically nest the right hotend into an available dock on the induction hotend rack.

⚠ DANGER

If the Nest operation is not performed from the printer screen, the induction hotend must be removed by gripping the nozzle tip. Since the nozzle has already been heated in Step 1, this poses a severe burn risk. Do not use this method.

- Step 3. On the printer touchscreen, tap **⚙ > Motion: XYZ**. Move the toolhead to a suitable position to enable easy removal of the induction hotend.
- Step 4. Remove the induction hotend from its corresponding dock. Hold only the heatsink section and avoid touching the nozzle tip.

**⚠ DANGER**

Do not touch the nozzle tip of a heated induction hotend to avoid burns.

- Step 5. Use pliers to hold the Allen key securely and heat it for approximately 10 seconds.
- Step 6. While the Allen key tip is still hot, quickly insert it vertically through the top of the heatsink into the hotend, pushing it into the clogged filament as shown below. Then wait